

ABHISHEK KUMAR

Data Science & Machine Learning Engineer — Intern / Fresher

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[LinkedIn](#) | [GitHub](#) | [Live Project: MindScore AI](#)

PROFESSIONAL SUMMARY

Final-year B.Tech Computer Science student (Data Science Specialization, 2027) with hands-on experience building and deploying end-to-end Machine Learning and Deep Learning systems. Skilled in Python, SQL, and ML pipelines using Scikit-learn, TensorFlow/Keras, Flask, and FastAPI. Delivered deployed, full-stack ML applications — a time-series forecasting model and a regression-based prediction API — covering data preprocessing, feature engineering, model tuning, and deployment. Seeking a Data Science / Machine Learning Engineer internship or fresher role to apply strong analytical, statistical, and problem-solving skills.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, Java
- **Machine Learning / Deep Learning:** Supervised Learning, Regression, Random Forest, LSTM (RNN), Model Evaluation & Hyperparameter Tuning (RandomizedSearchCV), Scikit-learn, TensorFlow, Keras
- **Data Analysis & Statistics:** EDA, Feature Engineering, Feature Scaling, Statistical Analysis, NumPy, Pandas, Matplotlib, Seaborn
- **Model Deployment & APIs:** Flask, FastAPI, Pydantic, REST APIs, Model Serialization (Pickle/Joblib), Render (Cloud Deployment)
- **Databases:** MySQL, MongoDB
- **Tools & Concepts:** Git, GitHub, Jupyter Notebook, Object-Oriented Programming (OOP), Database Management System.

PROJECTS

Mental Health Score Prediction System (MindScore AI) [\[Live Demo\]](#) [\[GitHub\]](#)

Python, Scikit-learn, Pandas, NumPy, FastAPI, Pydantic, Random Forest Regression, Render

- Built an end-to-end ML system predicting student mental health scores (0-10) from social-media, academic, sleep, and lifestyle data, covering EDA, data cleaning, feature engineering, and statistical analysis on a real-world survey dataset.
- Engineered a Scikit-learn preprocessing pipeline (ColumnTransformer) combining log-transformation, StandardScaler, OrdinalEncoder, and OneHotEncoder across 12 mixed feature types.
- Trained and benchmarked Linear Regression vs. Random Forest Regression; tuned the Random Forest via RandomizedSearchCV (5-fold CV) to achieve Test $R^2 = 0.878$, MAE = 0.347, outperforming the linear baseline ($R^2 = 0.740$) by 18.6%.
- Deployed the serialized model pipeline as a FastAPI REST API with strict Pydantic request validation, connected to a vanilla JS/HTML/CSS frontend with real-time animated result visualization; live-hosted on Render.

Stock Price Prediction System (LSTM Time-Series Forecasting) [\[GitHub\]](#)

Python, TensorFlow/Keras, LSTM, Pandas, NumPy, Matplotlib, Flask, yfinance

- Designed a stacked 2-layer LSTM neural network (30,651 parameters) to forecast next-day stock closing prices from 10 years of historical OHLCV data retrieved live via the yfinance API.
- Implemented a 60-day sliding-window sequence pipeline with Min-Max feature scaling and a chronological (non-shuffled) 80/20 train-test split to prevent data leakage in time-series evaluation.
- Achieved a Test RMSE of ≈ 7.82 on unseen price data across 30 training epochs with Adam optimizer, validated by plotting predicted vs. actual price curves; deployed via Flask with pre-loaded model/scaler for low-latency inference.

WORK EXPERIENCE

Virtual AI Intern — IBM SkillsBuild September 2025 (6 Weeks)

- Completed a structured 6-week virtual internship covering core Artificial Intelligence and Machine Learning fundamentals, including supervised/unsupervised learning concepts and Deep Learning applications.
- Applied AI/ML theory to practical exercises, strengthening foundational knowledge later used to independently build and deploy two full-stack ML projects.

EDUCATION

B.Tech in Computer Science and Engineering — Data Science Specialization

Lloyd Institute of Engineering and Technology, Greater Noida | 2023 – 2027 | CGPA: 7.3 / 10

CERTIFICATIONS & ADDITIONAL INFORMATION

- **Certification:** IBM SkillsBuild — Virtual AI Internship Certificate (September 2025)
- **Languages:** English (Fluent), Hindi (Native)
- **Availability:** Open for Data Science / ML Engineer Internships and Live Projects